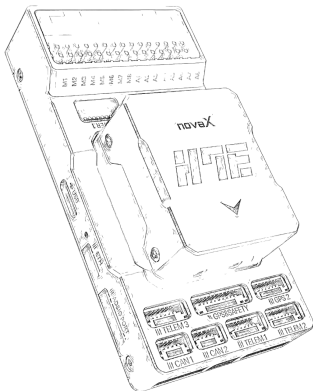


# AF-H7E

## User Guide



# AF-H7E Overview



Modular STM32H753 flight controller on the Pixhawk FMUv6x architecture with three IMUs, an RM3100 compass, two ICP-20100 barometers, 8 FMU + 8 IOMCU outputs, 2 CAN ports and Ethernet.

# AF-H7E Specifications

## • Basic Specifications

|                      |                                                                       |
|----------------------|-----------------------------------------------------------------------|
| MCU                  | STM32H753, ARM Cortex-M7,<br>480 MHz                                  |
| RAM / Flash          | 1 MB / 2 MB                                                           |
| IMU                  | BMI088 (Bosch, accel / gyro)                                          |
| Secondary IMU        | ICM-42688-P, ICM-20649<br>(TDK InvenSense)                            |
| IMU Heater           | Built-in heater on the IMU<br>board, held at 45 °C<br>(BRD_HEAT_TARG) |
| Magnetometer         | RM3100 (PNI)                                                          |
| Barometer            | 2× ICP-20100 (TDK<br>InvenSense)                                      |
| Operating<br>Voltage | 4.75 – 5.7 V (Rated 5 V)                                              |
| USB Input            | 4.75 – 5.25 V                                                         |
| Servo Rail           | 0 – 9.9 V                                                             |

# AF-H7E Specifications

## • Basic Specifications

|                  |                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| PWM Output       | 8 FMU + 8 IOMCU channels                                                                                                                       |
| RC Input         | S.Bus, PPM, DSM / Spektrum                                                                                                                     |
| RSSI Input       | Analog / PWM                                                                                                                                   |
| UART             | 7 peripheral serial interfaces + dedicated IOMCU link                                                                                          |
| Serial Mapping   | SERIAL1 = UART7 · SERIAL2 = UART5 · SERIAL3 = USART1 · SERIAL4 = UART8 · SERIAL5 = USART2 · SERIAL6 = UART4 · SERIAL7 = USART3 (USB = SERIAL0) |
| I <sup>2</sup> C | 4 configured buses including internal sensors; 3 exposed on multifunction ports                                                                |
| CAN              | 2 Port                                                                                                                                         |

# AF-H7E Specifications

## • Basic Specifications

|                |                                             |
|----------------|---------------------------------------------|
| ADC            | VBat/Current + Aux Analog Input             |
| Size           | 45 × 90 × 29.2 mm                           |
| Mount Hole     | Pixhawk FMUv6x Standard                     |
| Weight         | 99 g (Core 43g + Baseboard 56g)             |
| Operating Temp | -20 ~ +85 °C                                |
| Supported F/W  | novaX ArduPilot (Copter and Plane releases) |
| Ethernet       | 100 Mbps x 1 Port                           |

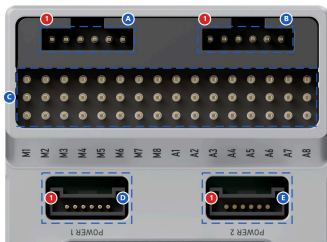
# AF-H7E Pinout

## • Pinout (1/2)

### AF-H7E Pinout — Top, Rear

Seen from above with the rear edge (PWM header) at the top. Each table reads left → right exactly as the picture next to it.

**A** = connector, its table below    **1** = pin 1, the red 1 in its table    + = 5 V    - = GND    blue = signal



#### **A** POWER C1

6-Pin - CAN power module - CAN1 bus - numbered in base-PCB pad order 0A → 0F

| 1 | 2 | 3     | 4     | 5 | 6 |
|---|---|-------|-------|---|---|
| - | - | CAN_L | CAN_H | + | + |



#### **B** POWER C2

6-Pin - CAN power module - CAN2 bus - numbered in base-PCB pad order 0G → 0L

| 1 | 2 | 3     | 4     | 5 | 6 |
|---|---|-------|-------|---|---|
| - | - | CAN_L | CAN_H | + | + |

# AF-H7E Pinout

## • Pinout (2/2)

### C PWM OUT — 3 × 16 header

M1–M8 = MAIN (IOMCU, SERVO1–8) · A1–A8 = AUX (FMU, SERVO9–16) · rows as seen: S rear, + middle, – front · + = servo rail, fed by an external BEC



| S | M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | A1 | A2 | A3 | A4 | A5 | A6 | A7 | A8 |
|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| + | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  | +  |
| - | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  | -  |



### D POWER 1

6-Pin · Micro-Lock Plus · primary power · I2C1 monitor

| 1 | 2 | 3   | 4   | 5 | 6 |
|---|---|-----|-----|---|---|
| + | + | SCL | SDA | - | - |



### E POWER 2

6-Pin · Micro-Lock Plus · redundant power · I2C2 monitor

| 1 | 2 | 3   | 4   | 5 | 6 |
|---|---|-----|-----|---|---|
| + | + | SCL | SDA | - | - |

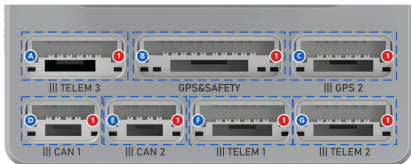
# AF-H7E Pinout

## • Pinout (1/2)

### AF-H7E Pinout — Top, Front Step

Seen from above with the front edge at the bottom. Each table reads left → right exactly as the picture next to it.

**A** = connector, its table below    **1** = pin 1, the red 1 in its table    + = 5 V    - = GND    blue = signal



#### **A** TELEM 3

SERIAL5 · 6-Pin · JST-GH

| 6 | 5   | 4   | 3  | 2  | 1 |
|---|-----|-----|----|----|---|
| - | RTS | CTS | RX | TX | + |



#### **B** GPS & SAFETY

SERIAL3 · 10-Pin · JST-GH

| 10 | 9      | 8   | 7          | 6         | 5   | 4   | 3  | 2  | 1 |
|----|--------|-----|------------|-----------|-----|-----|----|----|---|
| -  | BUZZER | 3V3 | SAFETY_LED | SAFETY_SW | SDA | SCL | RX | TX | + |



#### **C** GPS 2

SERIAL4 · 6-Pin · JST-GH

| 6 | 5   | 4   | 3  | 2  | 1 |
|---|-----|-----|----|----|---|
| - | SDA | SCL | RX | TX | + |

# AF-H7E Pinout

## • Pinout (2/2)



### D CAN 1

4-Pin - JST-GH

| 4 | 3     | 2     | 1 |
|---|-------|-------|---|
| - | CAN_L | CAN_H | + |



### E CAN 2

4-Pin - JST-GH

| 4 | 3     | 2     | 1 |
|---|-------|-------|---|
| - | CAN_L | CAN_H | + |



### F TELEM 1

SERIAL1 - 6-Pin - JST-GH

| 6 | 5   | 4   | 3  | 2  | 1 |
|---|-----|-----|----|----|---|
| - | RTS | CTS | RX | TX | + |



### G TELEM 2

SERIAL2 - 6-Pin - JST-GH

| 6 | 5   | 4   | 3  | 2  | 1 |
|---|-----|-----|----|----|---|
| - | RTS | CTS | RX | TX | + |

# AF-H7E Pinout

## • Pinout

### AF-H7E Pinout — Left Side

Seen from the left side; the front of the board is to the right. Each table reads left → right exactly as the picture next to it.

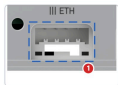
**A** = connector, its table below    **1** = pin 1, the red 1 in its table    + = 5 V    - = GND    blue = signal



#### **A** USB-C

SERIAL0 · Type-C · reversible — contacts in the pin table on this page

USB 2.0



#### **B** ETHERNET

4-Pin · JST-GH

|     |     |     |     |
|-----|-----|-----|-----|
| 4   | 3   | 2   | 1   |
| TX+ | TX- | RX+ | RX- |



#### **C** AD & IO

8-Pin · JST-GH

|   |         |         |        |         |            |      |   |
|---|---------|---------|--------|---------|------------|------|---|
| 8 | 7       | 6       | 5      | 4       | 3          | 2    | 1 |
| - | ADC_6V6 | ADC_3V3 | nARMED | RST_REQ | BOOTLOADER | CAP1 | + |

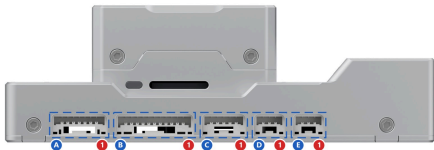
# AF-H7E Pinout

## • Pinout (1/2)

### AF-H7E Pinout — Right Side

Seen from the right side; the rear (PWM header) is to the right. Each table reads left → right exactly as the picture next to it.

**A** = connector, its table below    **1** = pin 1, the red 1 in its table    + = 5 V    - = GND    blue = signal



#### **A** UART 4

SERIAL6 · 7-Pin · JST-GH

| 7 | 6        | 5   | 4   | 3  | 2  | 1 |
|---|----------|-----|-----|----|----|---|
| - | NFC_GPIO | SDA | SCL | RX | TX | + |



#### **B** SPI 6

11-Pin · JST-GH

| 11 | 10     | 9     | 8     | 7    | 6   | 5   | 4    | 3    | 2   | 1 |
|----|--------|-------|-------|------|-----|-----|------|------|-----|---|
| -  | nRESET | DRDY2 | DRDY1 | SYNC | CS2 | CS1 | MOSI | MISO | SCK | + |



#### **C** DSM / SBUS RC

5-Pin · JST-GH

| 5 | 4   | 3    | 2     | 1 |
|---|-----|------|-------|---|
| - | 3V3 | RSSI | RC_IN | + |

# AF-H7E Pinout

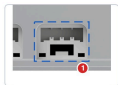
## • Pinout (2/2)



### D PPM IN

3-Pin - JST-GH

|   |     |   |
|---|-----|---|
| 3 | 2   | 1 |
| - | PPM | + |



### E SBUS OUT

3-Pin - JST-GH

|   |          |    |
|---|----------|----|
| 3 | 2        | 1  |
| - | SBUS_OUT | NC |

# AF-H7E Pinout

## • Pinout

### AF-H7E Pinout — Front End

Seen from the front end. Each table reads left → right exactly as the picture next to it.

**A** = connector, its table below    **1** = pin 1, the red 1 in its table    + = 5 V    - = GND    blue = signal



#### **A** FMU DEBUG

SERIAL7 · 10-Pin · JST-SH

| 10 | 9    | 8    | 7        | 6   | 5     | 4     | 3  | 2  | 1   |
|----|------|------|----------|-----|-------|-------|----|----|-----|
| -  | nRST | PH11 | NFC_GPIO | SWO | SWCLK | SWDIO | RX | TX | 3V3 |



#### **B** USB

4-Pin · JST-GH

| 4 | 3  | 2  | 1    |
|---|----|----|------|
| - | D+ | D- | VBUS |



#### **C** IO DEBUG

10-Pin · JST-SH

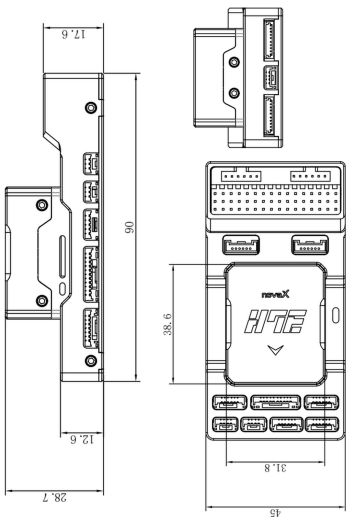
| 10 | 9    | 8     | 7     | 6   | 5     | 4     | 3  | 2  | 1   |
|----|------|-------|-------|-----|-------|-------|----|----|-----|
| -  | nRST | GPIO2 | GPIO1 | SWO | SWCLK | SWDIO | NC | TX | 3V3 |

# AF-H7E Pinout

## • Dimensions

### AF-H7E Dimensions

Millimetres. Side view (top), top view and front view (bottom).



# AF-H7E Pinout

## • Pinout Notes

I2C bus numbering on the multi-function ports: the GPS & Safety port carries I2C1, the GPS2 port carries I2C2, and the UART4 port carries I2C3.

# AF-H7E Pin Definition

## • POWER 1

Molex Micro-Lock Plus 6P (1.25 mm) · top, left of the FMU ·  
Primary power input · I2C1 power monitor

| Pin | Signal | Function                                           |
|-----|--------|----------------------------------------------------|
| 1   | VCC_IN | 5 V supply input from the power module             |
| 2   | VCC_IN | 5 V supply input from the power module             |
| 3   | SCL    | Power-module I2C clock (voltage / current monitor) |
| 4   | SDA    | Power-module I2C data (voltage / current monitor)  |
| 5   | GND    | Ground                                             |
| 6   | GND    | Ground                                             |

# AF-H7E Pin Definition

## • POWER 2

Molex Micro-Lock Plus 6P (1.25 mm) · top, right of the FMU ·  
Redundant power input · I2C2 power monitor

| Pin | Signal | Function                                           |
|-----|--------|----------------------------------------------------|
| 1   | VCC_IN | 5 V supply input from the second power module      |
| 2   | VCC_IN | 5 V supply input from the second power module      |
| 3   | SCL    | Power-module I2C clock (voltage / current monitor) |
| 4   | SDA    | Power-module I2C data (voltage / current monitor)  |
| 5   | GND    | Ground                                             |
| 6   | GND    | Ground                                             |

# AF-H7E Pin Definition

## • POWER C1

6P · top, rear left (on the PWM header block) · CAN power module input · CAN1

| Pin | Signal | Function                                   |
|-----|--------|--------------------------------------------|
| 1   | GND    | Ground                                     |
| 2   | GND    | Ground                                     |
| 3   | CAN_L  | CAN1 bus low (same bus as the CAN 1 port)  |
| 4   | CAN_H  | CAN1 bus high                              |
| 5   | VCC_IN | 5 V supply input from the CAN power module |
| 6   | VCC_IN | 5 V supply input from the CAN power module |

# AF-H7E Pin Definition

## • POWER C2

6P · top, rear right (on the PWM header block) · CAN power module input · CAN2

| Pin | Signal | Function                                          |
|-----|--------|---------------------------------------------------|
| 1   | GND    | Ground                                            |
| 2   | GND    | Ground                                            |
| 3   | CAN_L  | CAN2 bus low (same bus as the CAN 2 port)         |
| 4   | CAN_H  | CAN2 bus high                                     |
| 5   | VCC_IN | 5 V supply input from the second CAN power module |
| 6   | VCC_IN | 5 V supply input from the second CAN power module |

# AF-H7E Pin Definition

## • TELEM 1

JST-GH 6P · front, lower row · SERIAL1 · UART7 · default  
MAVLink2

| Pin | Signal | Function                                       |
|-----|--------|------------------------------------------------|
| 1   | VCC    | 5 V output                                     |
| 2   | TX     | UART transmit (FC → device)                    |
| 3   | RX     | UART receive (device → FC)                     |
| 4   | CTS    | Clear to send — hardware flow control input    |
| 5   | RTS    | Request to send — hardware flow control output |
| 6   | GND    | Ground                                         |

# AF-H7E Pin Definition

## • TELEM 2

JST-GH 6P · front, lower row · SERIAL2 · UART5 · default  
MAVLink2

| Pin | Signal | Function                                       |
|-----|--------|------------------------------------------------|
| 1   | VCC    | 5 V output                                     |
| 2   | TX     | UART transmit (FC → device)                    |
| 3   | RX     | UART receive (device → FC)                     |
| 4   | CTS    | Clear to send — hardware flow control input    |
| 5   | RTS    | Request to send — hardware flow control output |
| 6   | GND    | Ground                                         |

# AF-H7E Pin Definition

## • TELEM 3

JST-GH 6P · front, upper row · SERIAL5 · USART2

| Pin | Signal | Function                                       |
|-----|--------|------------------------------------------------|
| 1   | VCC    | 5 V output                                     |
| 2   | TX     | UART transmit (FC → device)                    |
| 3   | RX     | UART receive (device → FC)                     |
| 4   | CTS    | Clear to send — hardware flow control input    |
| 5   | RTS    | Request to send — hardware flow control output |
| 6   | GND    | Ground                                         |

# AF-H7E Pin Definition

## • GPS & SAFETY

JST-GH 10P · front, upper row · SERIAL3 · USART1 + I2C1 · default GPS

| Pin | Signal     | Function                               |
|-----|------------|----------------------------------------|
| 1   | VCC        | 5 V output                             |
| 2   | TX         | UART transmit (FC → GPS)               |
| 3   | RX         | UART receive (GPS → FC)                |
| 4   | SCL        | I2C1 clock — compass in the GPS module |
| 5   | SDA        | I2C1 data                              |
| 6   | SAFETY_SW  | Safety switch input                    |
| 7   | SAFETY_LED | Safety switch LED output               |
| 8   | VCC_3V3    | 3.3 V supply for the safety switch     |
| 9   | BUZZER     | Buzzer output                          |
| 10  | GND        | Ground                                 |

# AF-H7E Pin Definition

## • GPS 2

JST-GH 6P · front, upper row · SERIAL4 · UART8 + I2C2 · default GPS

| Pin | Signal | Function                               |
|-----|--------|----------------------------------------|
| 1   | VCC    | 5 V output                             |
| 2   | TX     | UART transmit (FC → GPS)               |
| 3   | RX     | UART receive (GPS → FC)                |
| 4   | SCL    | I2C2 clock — compass in the GPS module |
| 5   | SDA    | I2C2 data                              |
| 6   | GND    | Ground                                 |

# AF-H7E Pin Definition

## • UART 4

JST-GH 7P · right side, front · SERIAL6 · UART4 + I2C3

| Pin | Signal   | Function                    |
|-----|----------|-----------------------------|
| 1   | VCC      | 5 V output                  |
| 2   | TX       | UART transmit (FC → device) |
| 3   | RX       | UART receive (device → FC)  |
| 4   | SCL      | I2C3 clock                  |
| 5   | SDA      | I2C3 data                   |
| 6   | NFC_GPIO | General-purpose I/O         |
| 7   | GND      | Ground                      |

## • CAN 1

JST-GH 4P · front, lower row · CAN1 · DroneCAN

| Pin | Signal | Function                                          |
|-----|--------|---------------------------------------------------|
| 1   | VCC    | 5 V output to CAN peripherals (not a power input) |
| 2   | CAN_H  | CAN bus high                                      |
| 3   | CAN_L  | CAN bus low                                       |
| 4   | GND    | Ground                                            |

# AF-H7E Pin Definition

## • CAN 2

JST-GH 4P · front, lower row · CAN2 · DroneCAN

| Pin | Signal | Function                                          |
|-----|--------|---------------------------------------------------|
| 1   | VCC    | 5 V output to CAN peripherals (not a power input) |
| 2   | CAN_H  | CAN bus high                                      |
| 3   | CAN_L  | CAN bus low                                       |
| 4   | GND    | Ground                                            |

## • DSM / SBUS RC

JST-GH 5P · right side · RC input through the IOMCU · SBUS / DSM

| Pin | Signal  | Function                                            |
|-----|---------|-----------------------------------------------------|
| 1   | VCC     | 5 V receiver supply                                 |
| 2   | RC_IN   | SBUS / DSM receiver input                           |
| 3   | RSSI    | Analog RSSI input                                   |
| 4   | VCC_3V3 | 3.3 V supply for DSM / Spektrum satellite receivers |
| 5   | GND     | Ground                                              |

# AF-H7E Pin Definition

## • PPM IN

JST-GH 3P · right side · PPM receiver input

| Pin | Signal | Function            |
|-----|--------|---------------------|
| 1   | VCC    | 5 V receiver supply |
| 2   | PPM    | PPM receiver input  |
| 3   | GND    | Ground              |

## • SBUS OUT

JST-GH 3P · right side, rear · SBUS output from the IOMCU

| Pin | Signal   | Function          |
|-----|----------|-------------------|
| 1   | NC       | Not connected     |
| 2   | SBUS_OUT | SBUS servo output |
| 3   | GND      | Ground            |

# AF-H7E Pin Definition

## • PWM OUT M1–M8

3 × 16 pin header · rows about 2.54 mm (servo plug S · + · -), channels 2.6 mm apart on the base PCB · columns 1–8 (left) · MAIN outputs from the IOMCU · SERVO1–8

| Pin | Signal    | Function                                                                          |
|-----|-----------|-----------------------------------------------------------------------------------|
| S   | M1 ... M8 | Motor / servo output signal, one per channel (rear row)                           |
| +   | V_SERVO   | Servo rail 0 – 9.9 V, bussed across all channels — supplied externally (e.g. BEC) |
| -   | GND       | Ground (front row)                                                                |

## • PWM OUT A1–A8

3 × 16 pin header · rows about 2.54 mm (servo plug S · + · -), channels 2.6 mm apart on the base PCB · columns 9–16 (right) · AUX outputs from the FMU · SERVO9–16

| Pin | Signal    | Function                                                                          |
|-----|-----------|-----------------------------------------------------------------------------------|
| S   | A1 ... A8 | Motor / servo output signal, one per channel (rear row)                           |
| +   | V_SERVO   | Servo rail 0 – 9.9 V, bussed across all channels — supplied externally (e.g. BEC) |
| -   | GND       | Ground (front row)                                                                |

# AF-H7E Pin Definition

## • ETHERNET

JST-GH 4P · left side · 100BASE-T

| Pin | Signal | Function                 |
|-----|--------|--------------------------|
| 1   | RX-    | Ethernet receive pair -  |
| 2   | RX+    | Ethernet receive pair +  |
| 3   | TX-    | Ethernet transmit pair - |
| 4   | TX+    | Ethernet transmit pair + |

## • USB-C

USB Type-C · left side, rear · SERIAL0 · USB 2.0 Full Speed

| Pin              | Signal    | Function                                                      |
|------------------|-----------|---------------------------------------------------------------|
| A4 B4<br>A9 B9   | VBUS      | 5 V from USB — configuration and firmware update on the bench |
| A6 B6            | D+        | USB data +                                                    |
| A7 B7            | D-        | USB data -                                                    |
| A5 B5            | CC1 / CC2 | Configuration channel (device role)                           |
| A1 B1<br>A12 B12 | GND       | Ground                                                        |

# AF-H7E Pin Definition

## • USB

JST-GH 4P · front end, middle · USB 2.0 — same data lines as the USB-C port (use one at a time)

| Pin | Signal | Function       |
|-----|--------|----------------|
| 1   | VBUS   | 5 V USB supply |
| 2   | D-     | USB data -     |
| 3   | D+     | USB data +     |
| 4   | GND    | Ground         |

## • AD & IO

JST-GH 8P · left side, front · FMU auxiliary inputs and outputs

| Pin | Signal         | Function                          |
|-----|----------------|-----------------------------------|
| 1   | VCC            | 5 V output                        |
| 2   | CAP1           | FMU timer capture input           |
| 3   | BOOTLOAD<br>ER | FMU bootloader request input      |
| 4   | RST_REQ        | FMU reset request input           |
| 5   | nARMED         | Armed status output (low = armed) |
| 6   | ADC_3V3        | Analog input, 0 – 3.3 V           |
| 7   | ADC_6V6        | Analog input, 0 – 6.6 V (divided) |
| 8   | GND            | Ground                            |

# AF-H7E Pin Definition

## • SPI 6

JST-GH 11P · right side · External SPI6 — not enabled in novaX firmware

| Pin | Signal | Function           |
|-----|--------|--------------------|
| 1   | VCC    | 5 V output         |
| 2   | SCK    | SPI clock          |
| 3   | MISO   | SPI data in        |
| 4   | MOSI   | SPI data out       |
| 5   | CS1    | Chip select 1      |
| 6   | CS2    | Chip select 2      |
| 7   | SYNC   | Sync output        |
| 8   | DRDY1  | Data-ready input 1 |
| 9   | DRDY2  | Data-ready input 2 |
| 10  | nRESET | Reset output       |
| 11  | GND    | Ground             |

# AF-H7E Pin Definition

## • FMU DEBUG

JST-SH 10P · front end, left · Pixhawk Debug Full · SERIAL7 ·  
USART3 · console + SWD

| Pin | Signal     | Function                                                |
|-----|------------|---------------------------------------------------------|
| 1   | VCC_3V3    | 3.3 V reference for the debug probe                     |
| 2   | CONSOLE_TX | Debug console transmit (FC → probe)                     |
| 3   | CONSOLE_RX | Debug console receive (probe → FC)                      |
| 4   | SWDIO      | SWD data — bootloader programming and firmware recovery |
| 5   | SWCLK      | SWD clock                                               |
| 6   | SWO        | Trace output (shared with SPI6 SCK)                     |
| 7   | NFC_GPIO   | General-purpose I/O                                     |
| 8   | PH11       | General-purpose I/O                                     |
| 9   | nRST       | FMU reset                                               |
| 10  | GND        | Ground                                                  |

# AF-H7E Pin Definition

## • IO DEBUG

JST-SH 10P · front end, right · IOMCU console + SWD

| Pin | Signal  | Function                            |
|-----|---------|-------------------------------------|
| 1   | VCC_3V3 | 3.3 V reference for the debug probe |
| 2   | IO_TX   | IOMCU console transmit              |
| 3   | NC      | Not connected                       |
| 4   | SWDIO   | IOMCU SWD data                      |
| 5   | SWCLK   | IOMCU SWD clock                     |
| 6   | SWO     | IOMCU trace output                  |
| 7   | GPIO1   | IOMCU spare I/O                     |
| 8   | GPIO2   | IOMCU spare I/O                     |
| 9   | nRST    | IOMCU reset                         |
| 10  | GND     | Ground                              |

# AF-H7E Setup

1. **Mounting** — Align the board's forward direction with the vehicle nose on a vibration-damped mount; otherwise set `AHRS_ORIENTATION`. Mounting: Pixhawk FMUv6x Standard.
2. **Power** — Supply power within the operating range and check polarity first. Operating voltage: 4.75 – 5.7 V (Rated 5 V).
3. **Peripherals** — Connect the receiver, GPS / compass and ESCs according to the pinout.
4. **Firmware** — Load this product's novaX ArduPilot image from the catalog Web Updater; never use another board's firmware.
5. **Setup** — In Mission Planner or similar, choose the frame, calibrate accelerometer, compass and radio, and set flight modes and failsafes.

# AF-H7E Firmware

## • Available Firmware

| Type                                                         | Version | Date       |
|--------------------------------------------------------------|---------|------------|
| ArduPilot Copter (.apj package)                              | 1.3.0   | 2026-08-20 |
| ArduPilot Copter - Bootloader + App (merged HEX / DFU / SWD) | 1.3.0   | 2026-08-20 |
| ArduPilot Plane (.apj package)                               | 1.3.0   | 2026-08-20 |
| ArduPilot Plane - Bootloader + App (merged HEX / DFU / SWD)  | 1.3.0   | 2026-08-20 |

Download files from the product page (QR).

# AF-H7E Safety Notes

- Always remove propellers while wiring, configuring or updating firmware.
- Pull connectors by the housing, never by the wires.
- Beware of static discharge; do not operate in wet or dusty conditions.
- Supply the servo rail (0 – 9.9 V) from an external BEC.
- Copter and Plane share board ID 6202 — pick the image by file name.



<Detailed Manual>

